

SIMATS ENGINEERING

ASSESSMENT TOOL II – INDUSTRY PROBLEM TASK

Course: Code: CO: CO5

Duration: Total Marks: 100

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COURSE OUTCOME COVERED

CO5: Analyze vulnerabilities in Windows and Linux systems and evaluate security weaknesses in messaging, web, and mobile applications using appropriate exploitation techniques and hacking tools. (BT5)

1. Windows Domain Network

Given:

- Total systems = 150
- Anonymous enumeration = 45%
- SMB vulnerable among them = 60%
- Each compromised system attacks 3 systems
- Success probability = 0.3

Initial stage:

45% of 150:

So, approximately 68 systems allow anonymous enumeration.

SMB-susceptible systems:

Initial compromised systems = 41

Second stage:

Each of 41 systems attempts 3 connections:

Expected successful compromises:

So, approximately 37 additional systems may be compromised.

Total:

78 out of 150 systems:

Conclusion: The estimated compromise is 52%, which is below the 70% large-scale compromise threshold. However, the risk is still high because lateral movement can rapidly increase the number of affected systems.

Priority: Weak/misconfigured shared folders should be prioritized, because they directly enable unauthorized access and lateral movement. NULL sessions and exposed RPC should also be mitigated.

2. RPC Services and Exploitation

Given:

- Total systems = 120
- RPC-exposed systems = 70
- Vulnerable = 40%
- Exploitation success = 0.5
- Detection probability = 0.25

Vulnerable systems:

So, 28 systems are vulnerable.

Expected successful compromises:

Therefore, 14 systems are expected to be compromised.

Expected detection alerts:

Therefore, approximately 7 detection alerts are expected.

Conclusion: An attacker trying to remain stealthy may avoid direct exploitation because there is a 25% detection probability per attempt, which can generate several alerts. Alternative strategies such as credential abuse or less detectable access methods may be considered. From the defender's perspective, RPC exposure should be reduced and IDS monitoring should be strengthened.

3. Cloud Linux Server

Given vulnerability probabilities:

- Service 1 = 0.3
- Service 2 = 0.2
- Service 3 = 0.25
- Service 4 = 0.15
- Weak permissions exploitation = 0.5

Probability of at least one vulnerable service

First calculate the probability that none of the services are vulnerable:

Therefore:

So the probability is 64.3%.

Overall probability of full compromise

Considering privilege escalation probability of 0.5:

Therefore, the overall probability is 32.15%.

Conclusion: The server has a significant risk because there is a 64.3% chance of at least one vulnerable service. Strengthening file permissions would reduce the final compromise probability, but it would not eliminate the initial service vulnerability risk. Therefore, both exposed services and file permissions should be secured.

4. NFS Unauthorized Access

Given:

- Subnet 1 = 0.4
- Subnet 2 = 0.3
- Subnet 3 = 0.2
- Privilege escalation = 0.7

At least one subnet gains access

Probability that none gains access:

Therefore:

So, probability of unauthorized access is 66.4%.

Complete system compromise

Therefore:

Conclusion: There is a 66.4% chance of unauthorized NFS access and a 46.48% chance of complete compromise.

Restricting NFS access to only one subnet would reduce the number of possible entry points. For example, if the organization permits only the first subnet:

Thus, restricting access to a single trusted subnet can significantly reduce the risk.

5. SQL Injection Attack

Given:

- Total records = 50,000
- Sensitive records = 30%
- Extraction rate = 500 records/second
- Firewall blocks after 40 seconds

Records extracted

So, 20,000 records can be extracted.

Percentage of data compromised

Therefore, 40% of the database could be compromised.

Sensitive records exposed

Sensitive records represent 15,000 records.

Assuming the extracted records are representative of the database:

So, approximately 6,000 sensitive records could be exposed.

Conclusion

The breach is high severity because an attacker can extract 20,000 records, or 40% of the database, within only 40 seconds. The firewall is useful because it limits the attack, but it is not sufficient by itself. The organization should also use parameterized queries, input validation, least-privilege database accounts and continuous monitoring to prevent SQL injection.